2.7 The periodic table — groups 2 and 7

Students will be assessed on their ability to:

1 Properties down group 2

a explain the trend in the first ionisation energy down group 2

b recall the reaction of the elements in group 2 with oxygen, chlorine and water

c recall the reactions of the oxides of group 2 elements with water and dilute acid, and their hydroxides with dilute acid

d recall the trends in solubility of the hydroxides and sulfates of group 2 elements

e recall the trends in thermal stability of the nitrates and the carbonates of the elements in groups 1 and 2 and explain these in terms of size and charge of the cations involved

f recall the characteristic flame colours formed by group 1 and 2 compounds and explain their origin in terms of electron transitions

g describe and carry out the following:

i experiments to study the thermal decomposition of group 1 and 2 nitrates and carbonates

ii flame tests on compounds of group 1 and 2

iii simple acid-base titrations using a range of indicators, acids and alkalis, to calculate solution concentrations in g dm^{-3} and mol dm^{-3}, e.g. measuring the residual alkali present after skinning fruit with potassium hydroxide

h demonstrate an understanding of how to minimise the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result.